VOLTAGE-DIVIDER BIAS

JFET DC BIASING

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORTORY

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

Voltage-Divider Bias

- Gate-to-Source Loop
- Drain-to-Source Loop
- Transconductance Curve

VOLTAGE-DIVIDER BIAS

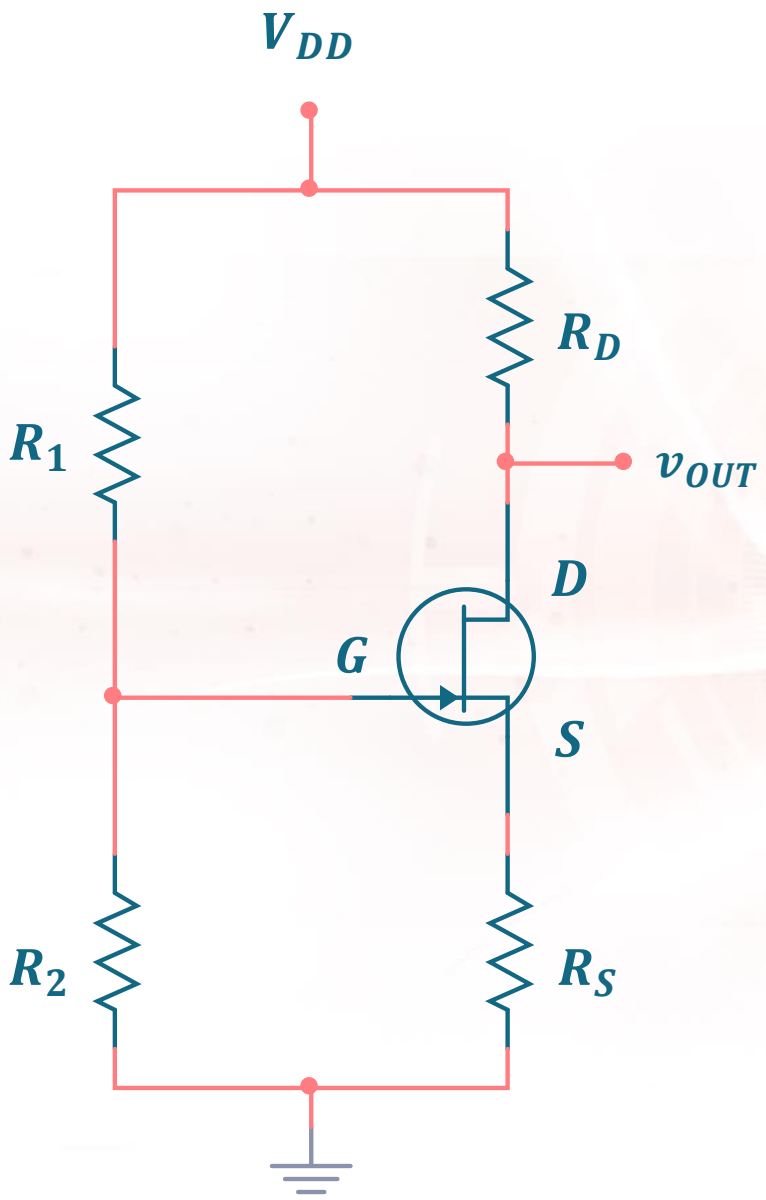
GENERAL RELATIONSHIPS

$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2$$

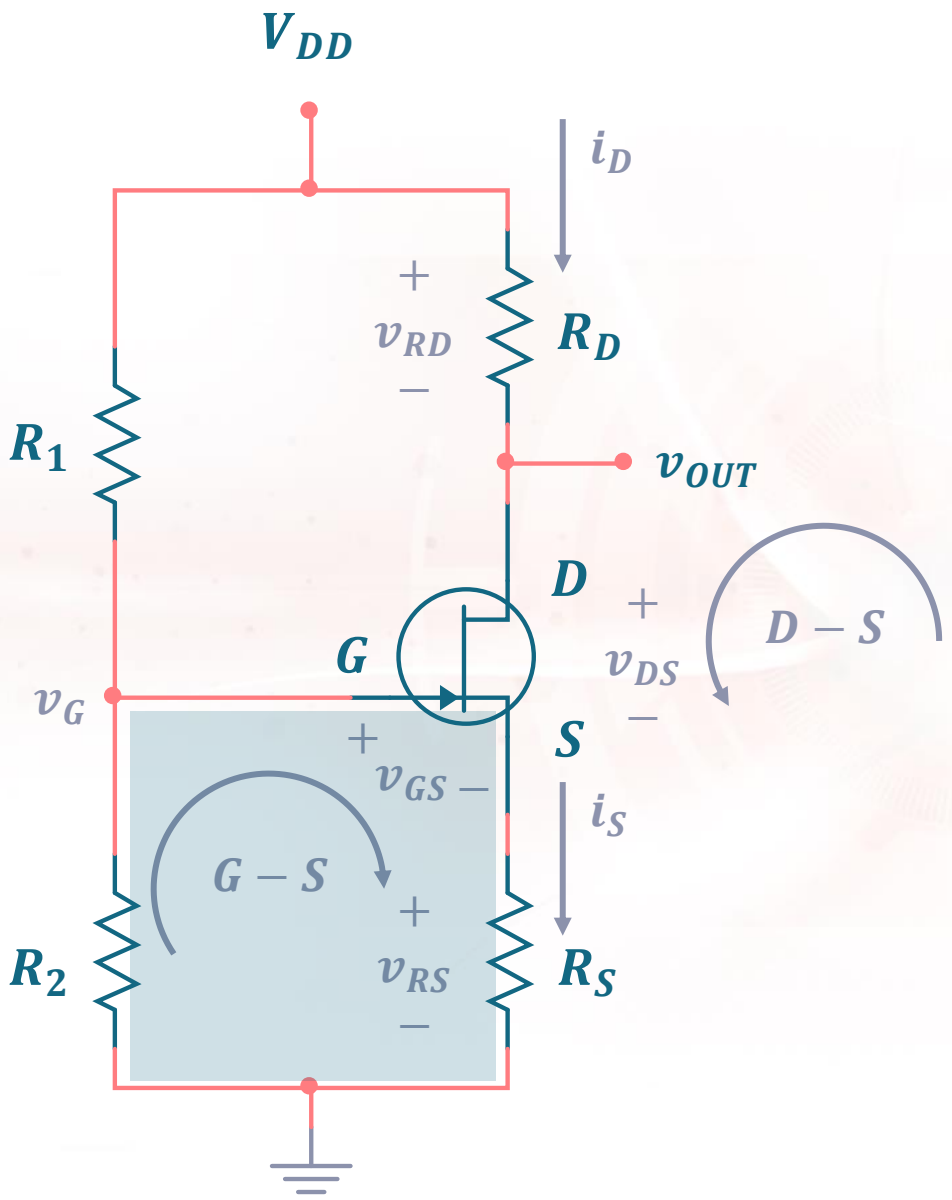
$$i_G \cong 0$$

$$i_D = i_S$$

VOLTAGE-DIVIDER BIAS CONFIGURATION



GATE-TO-SOURCE



KVL @ G-S

$$-v_G + v_{GS} + v_{RS} = 0$$

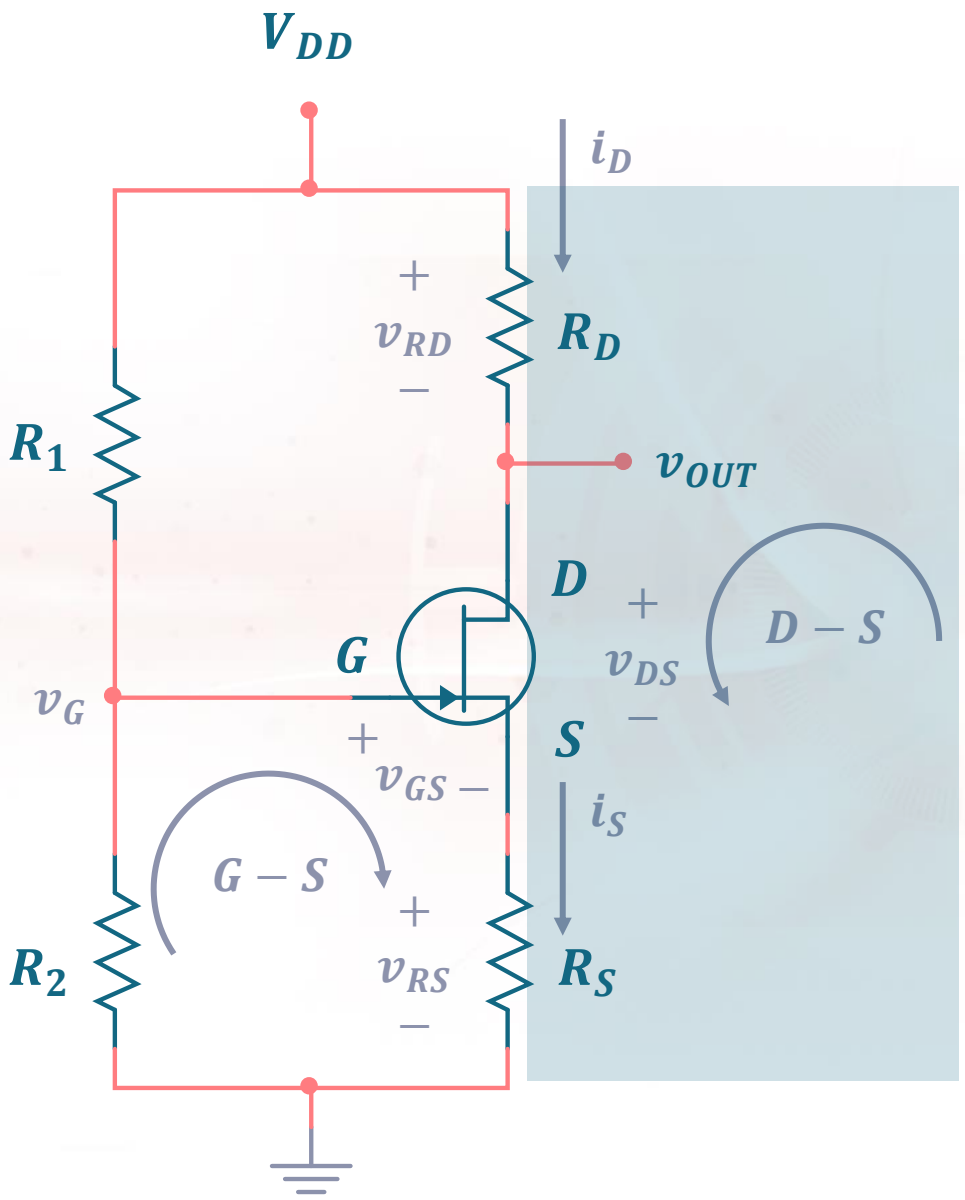
$$v_{GS} = v_G - v_{RS}$$

$$\boxed{v_{GS} = v_G - i_D R_S}$$

by VDT,

$$v_G = v_{DD}\frac{R_2}{R_1 + R_2}$$

DRAIN-TO-SOURCE



KVL @ D-S

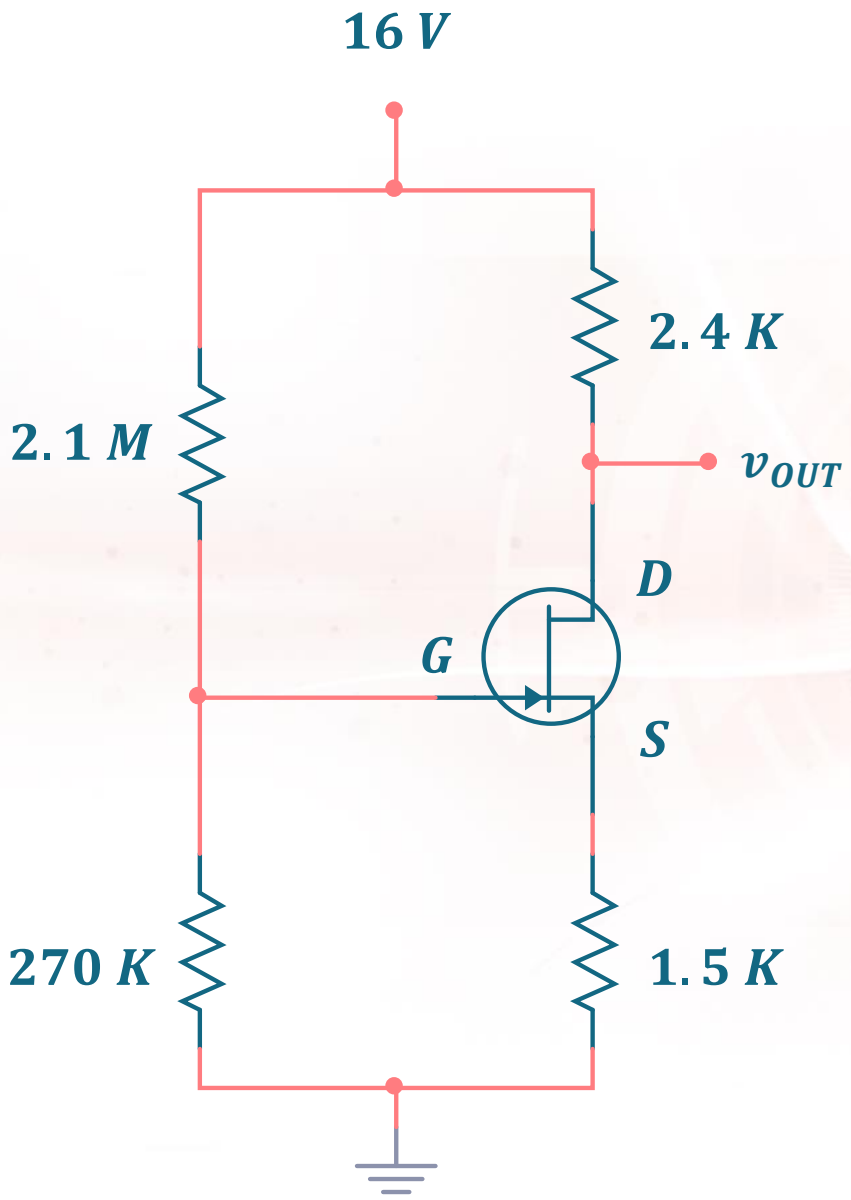
$$-v_{DD} + v_{RD} + v_{DS} + v_{RS} = 0$$

$$v_{DS} = v_{DD} - v_{RD} - v_{RS}$$

$$v_{DS} = v_{DD} - i_D R_D - i_S R_S$$

$$\boxed{v_{DS} = v_{DD} - i_D(R_D + R_S)}$$

EXERCISE



For the given network, determine the following :

v_G

v_S

v_{GSQ}

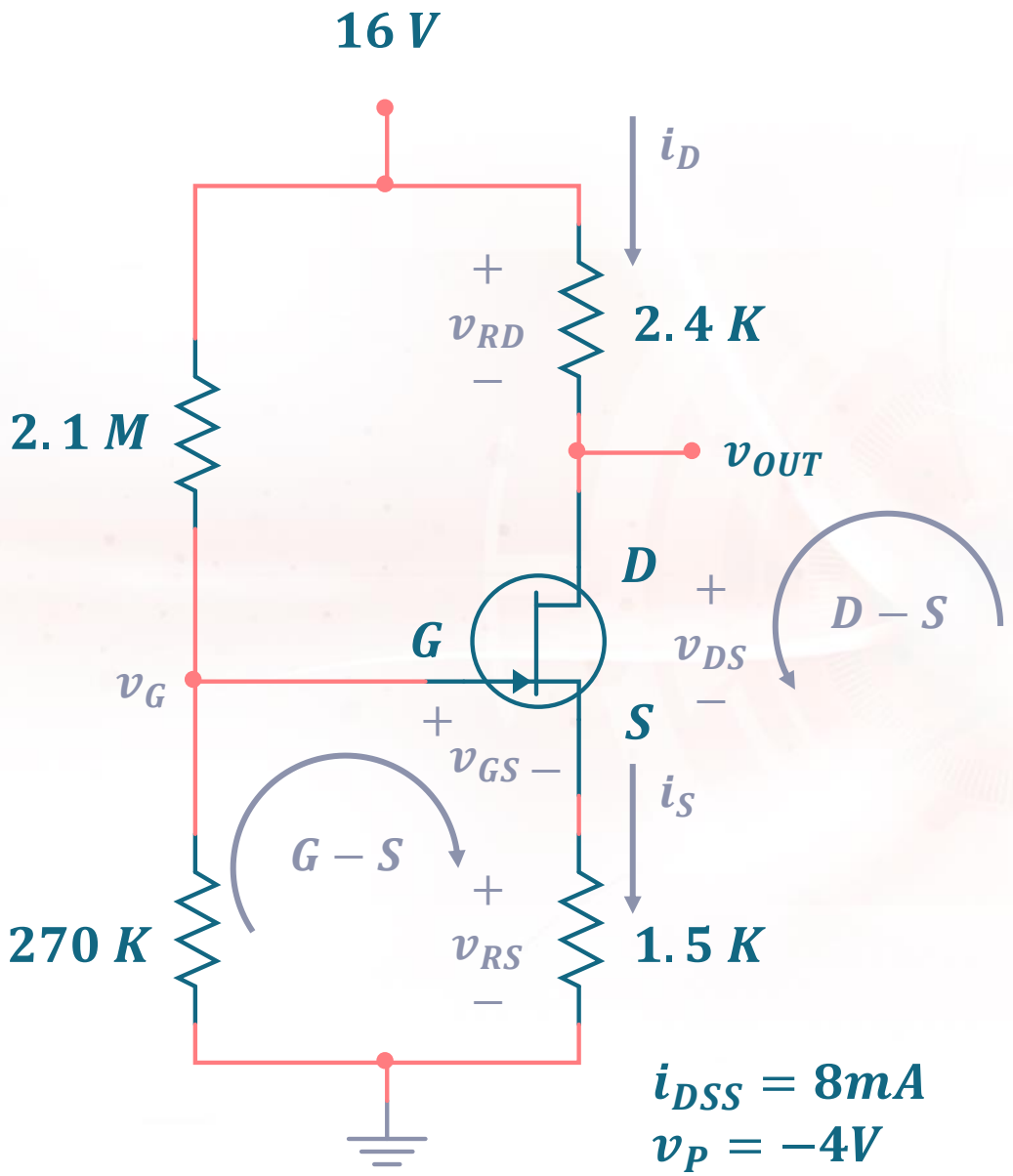
i_{DQ}

v_{DS}

v_S

and **sketch** the transconductance curve.

EXERCISE



$i_{DSS} = 8mA$
$v_P = -4V$

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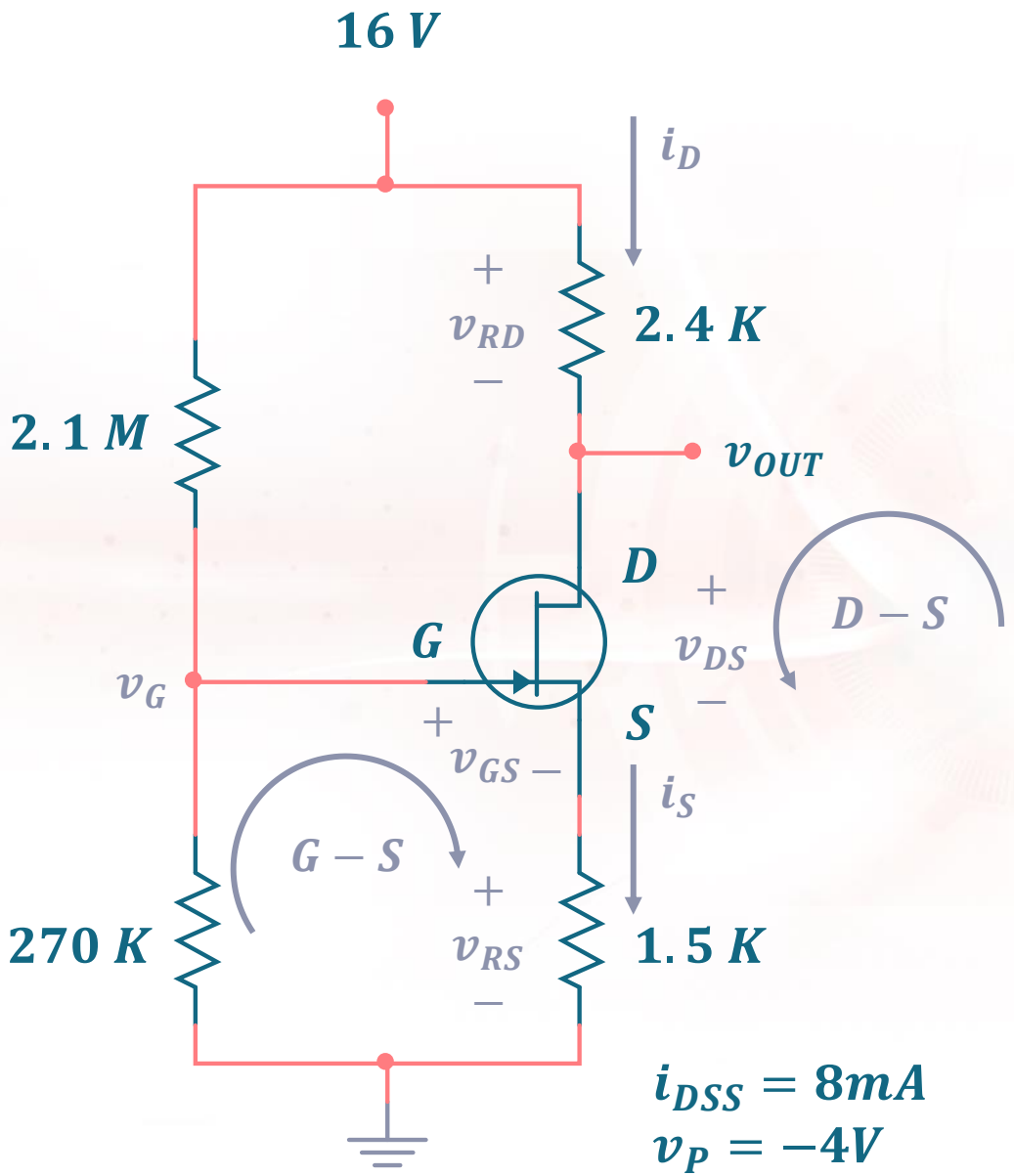
and **sketch** the transconductance curve.



$i_{DSS} = 8mA$
$v_P = -4V$

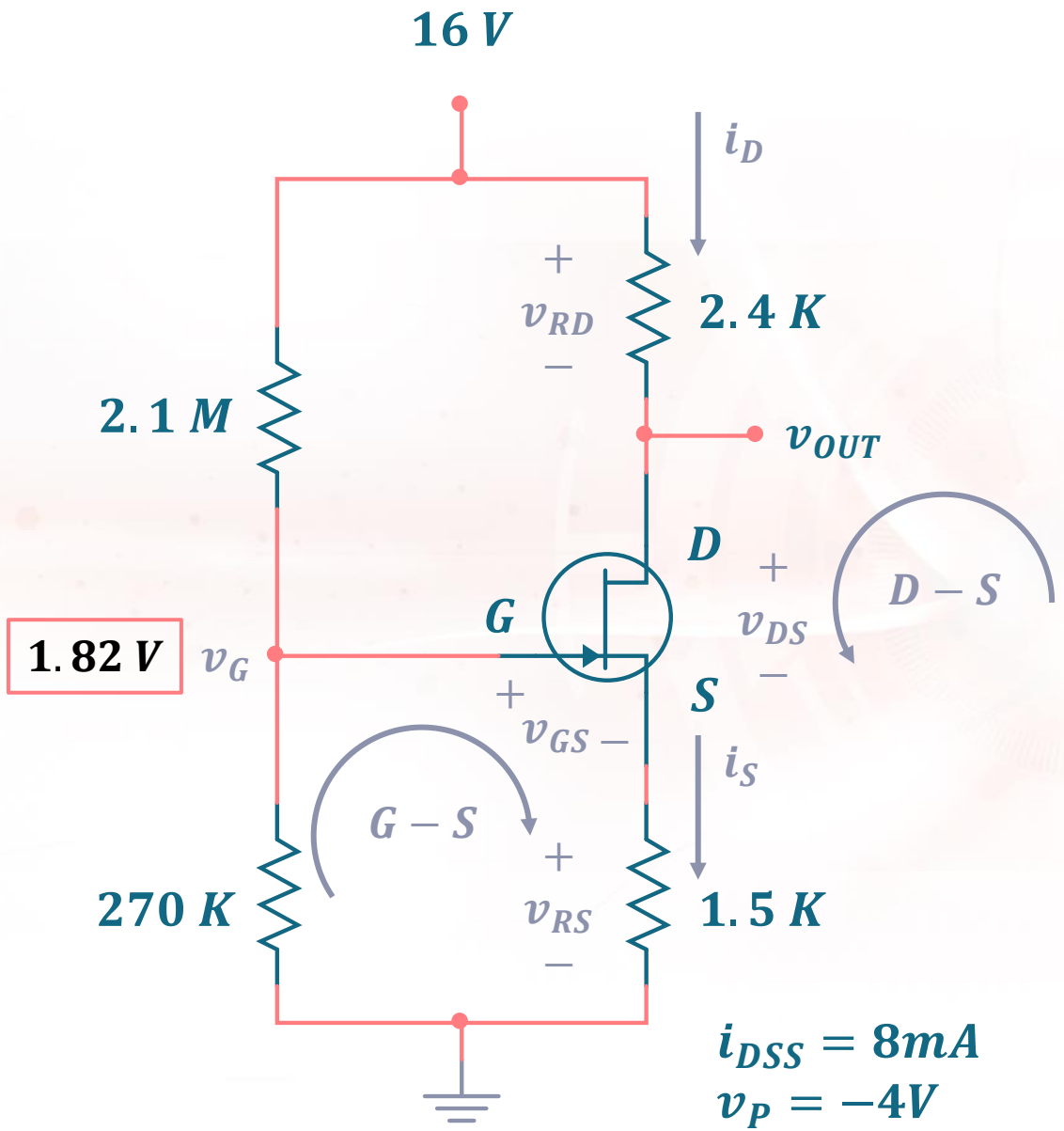
Gate Voltage

$$v_G = v_{DD}\frac{R_2}{R_1 + R_2}$$

$$v_G = 16\left(\frac{270\ K}{2.1\ M + 270\ K}\right)$$

$$\boxed{v_G = 1.82\ V}$$

ans



$i_{DSS} = 8mA$
$v_P = -4V$

KVL @ G-S

$$-v_G + v_{GS} + v_{RS} = 0$$

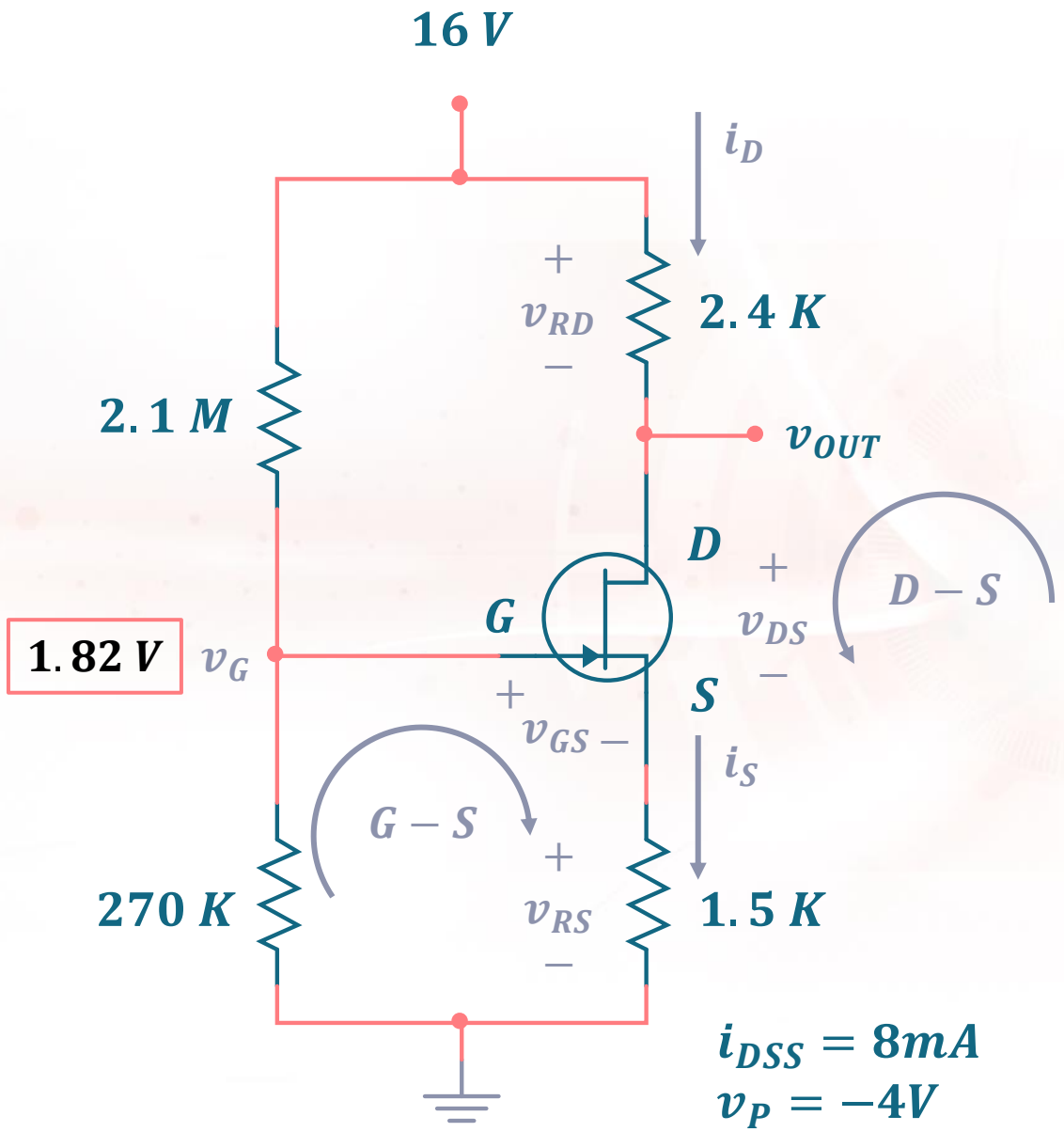
$$v_{GS} = v_G - v_{RS}$$

$$v_{GS} = v_G - i_D R_S$$

$$v_{GS} = v_G - i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2 R_S$$

$$v_{GS} = 1.82 - 8\,mA\left(1 - \frac{v_{GS}}{-4}\right)^2 1.5\,K$$

$$v_{GS} = 1.82 - 12\left(1 + \frac{1}{2}v_{GS} + \frac{v_{GS}^2}{16}\right)$$



$i_{DSS} = 8mA$
$v_P = -4V$

KVL @ G-S

$$v_{GS} = 1.82 - 12\left(1 + \frac{1}{2}v_{GS} + \frac{v_{GS}^2}{16}\right)$$

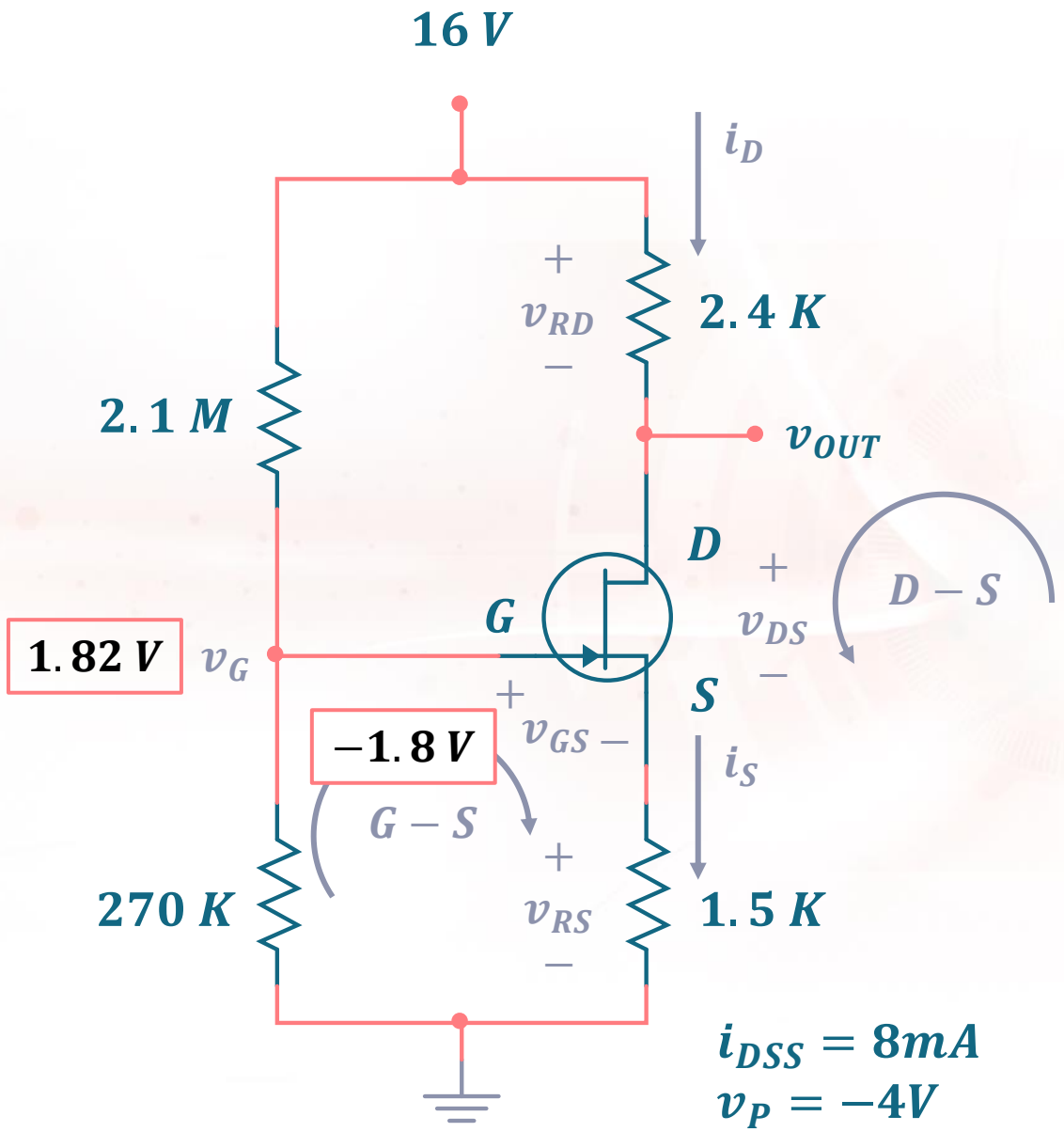
$$v_{GS} = 1.82 - 12 - 6v_{GS} - 0.75v_{GS}^2$$

$$0 = -10.18 - 7v_{GS} - 0.75v_{GS}^2$$

$$v_{GS} = -7.53\ V$$

$$\boxed{v_{GS} = -1.8\ V}$$

ans



$i_{DSS} = 8mA$
$v_P = -4V$

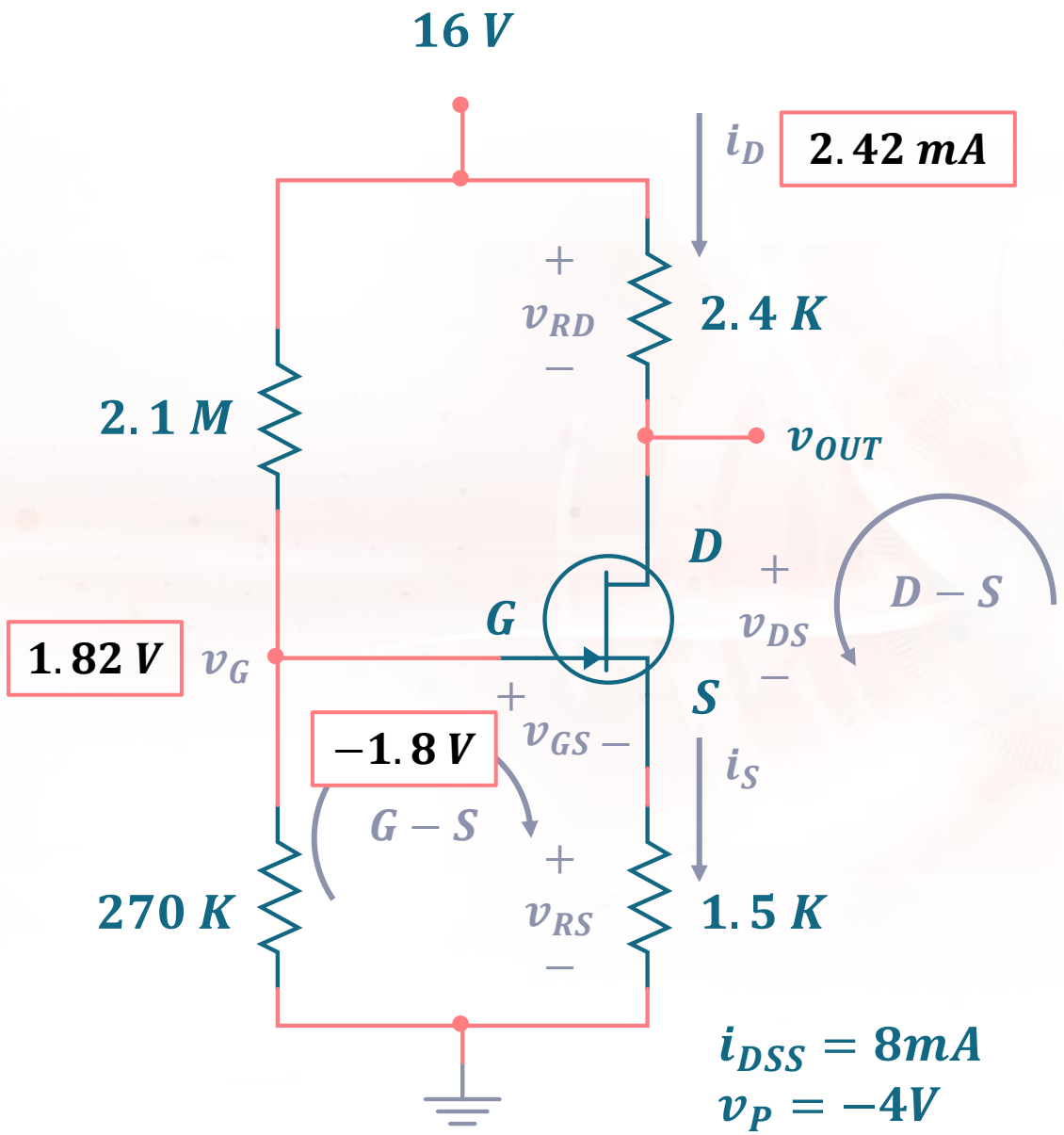
Drain Current

$$i_D = i_{DSS}\left(1 - \frac{v_{GS}}{v_P}\right)^2$$

$$i_D = 8\ mA\left(1 - \frac{-1.8}{-4}\right)^2$$

$$\boxed{i_{DQ} = 2.42\ mA}$$

ans



$i_{DSS} = 8mA$
$v_P = -4V$

KVL @ D-S

$$-v_{DD} + v_{RD} + v_{DS} + v_{RS} = 0$$

$$v_{DS} = v_{DD} - v_{RD} - v_{RS}$$

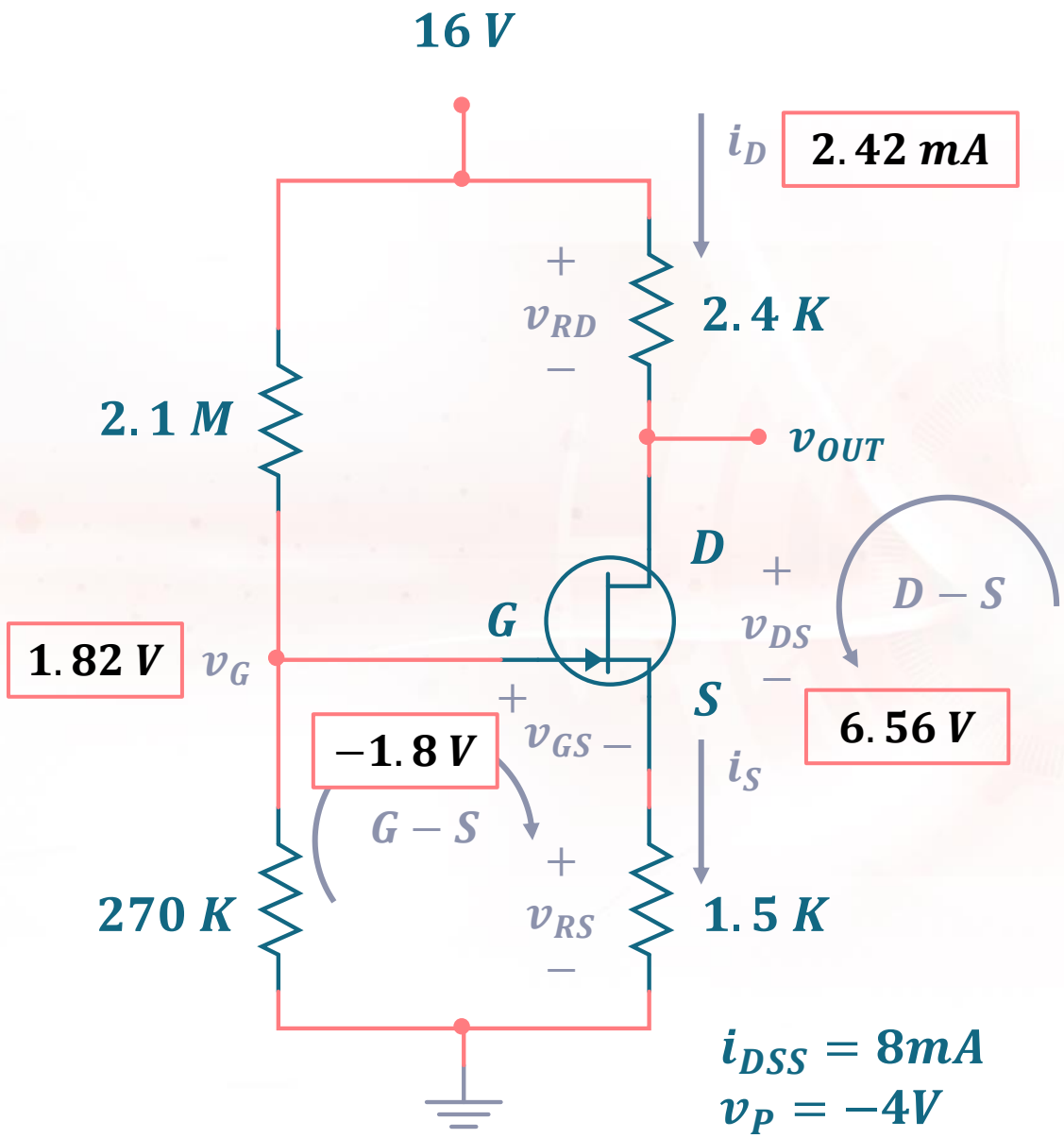
$$v_{DS} = v_{DD} - i_D R_D - i_S R_S$$

$$v_{DS} = v_{DD} - i_D(R_D + R_S)$$

$$v_{DS} = 16 - 2.42\ mA(2.4\ K + 1.5\ K)$$

$$\boxed{v_{DS} = 6.56\ V}$$

ans



$i_{DSS} = 8mA$
$v_P = -4V$

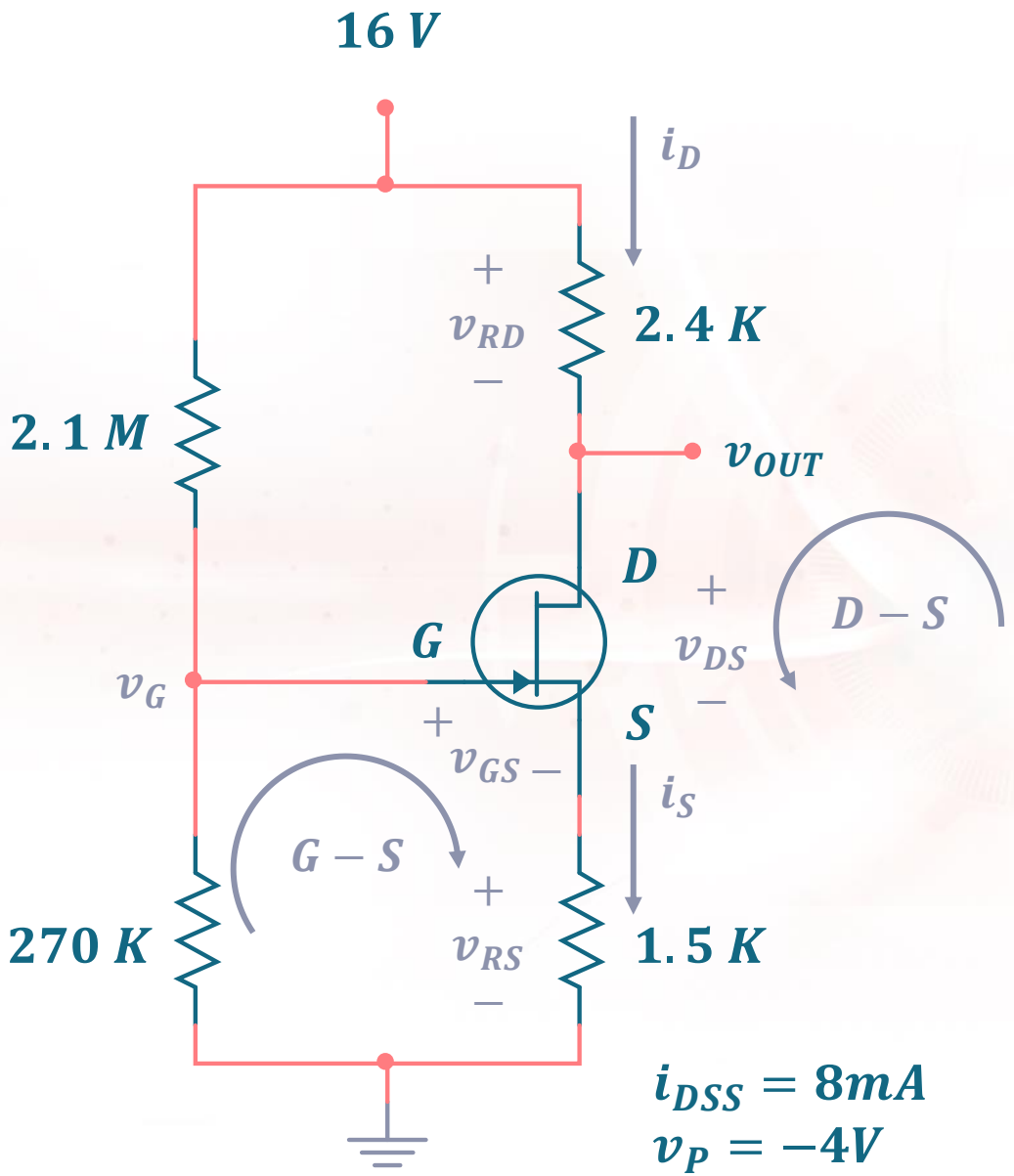
Source Voltage

$$v_S = i_D R_S$$

$$v_S = 2.42\ mA(1.5\ K)$$

$$\boxed{v_S = 3.63\ V}$$

ans



$i_{DSS} = 8mA$
$v_P = -4V$

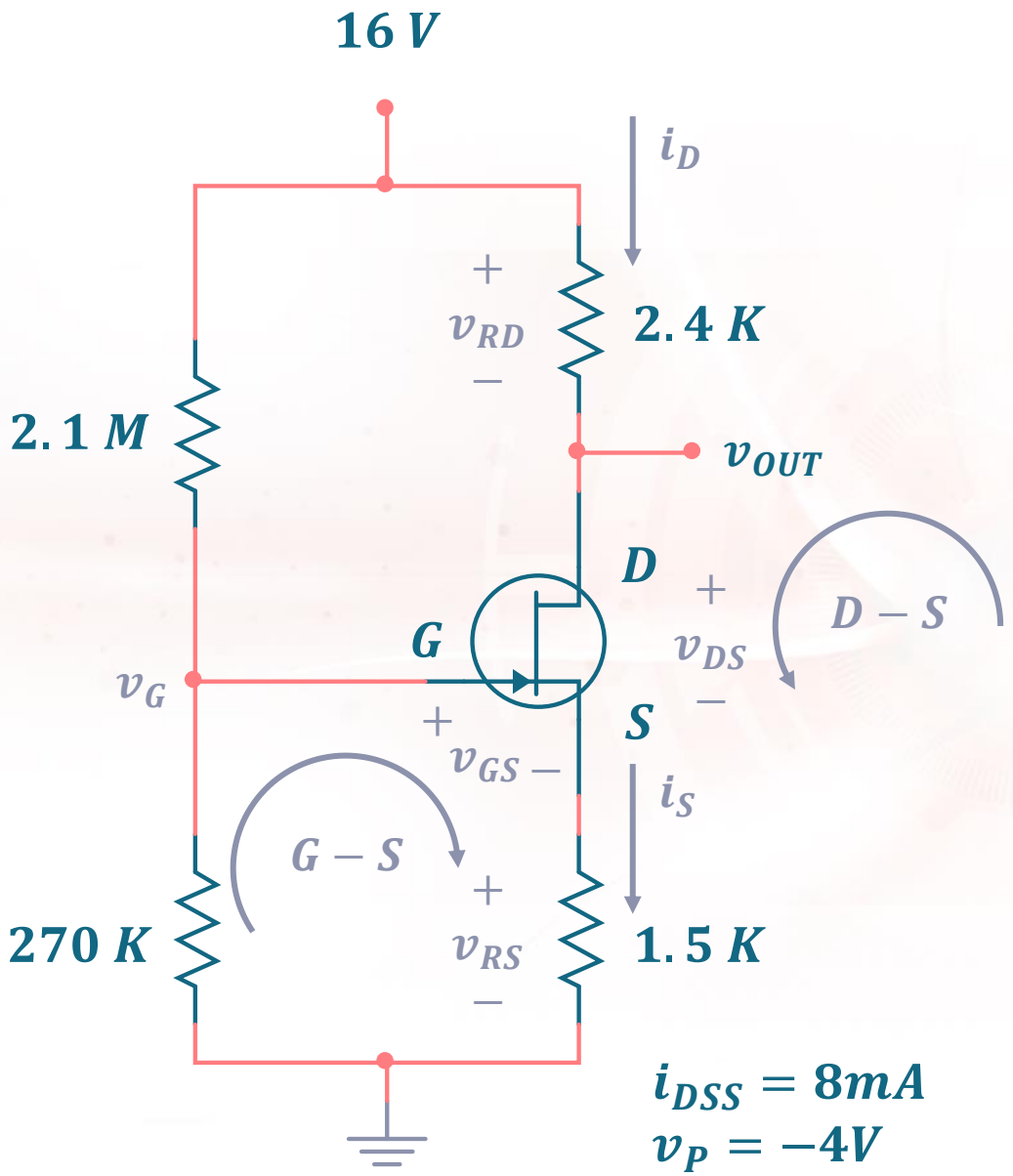
Transconductance Curve

$$i_D = \frac{1}{4} i_{DSS}\Big|_{v_{GS}=0.5v_P}$$

$$i_D = \frac{1}{4}(8\ mA)\Big|_{v_{GS}=-2\ V}$$

$$\boxed{i_D = 2\ mA\Big|_{v_{GS}=-2\ V}}$$

ans



$i_{DSS} = 8mA$
$v_P = -4V$

Transconductance Curve

$$v_{GS} = 0.3v_P\Big|_{i_d=0.5i_{DSS}}$$

$$v_{GS} = 0.3(-4)\Big|_{i_D=4\ mA}$$

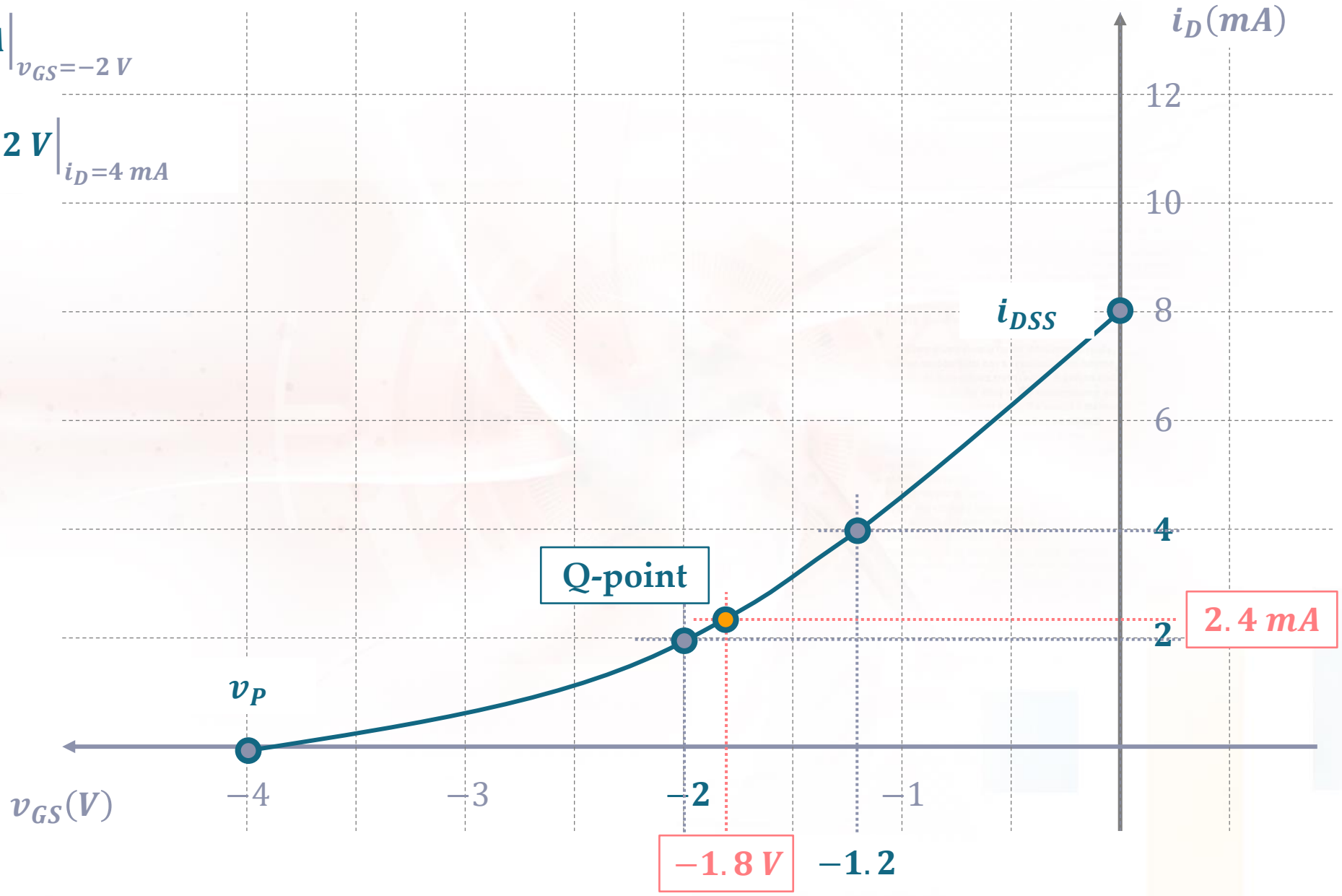
$$\boxed{v_{GS} = -1.2\ V\Big|_{i_D=4\ mA}}$$

ans

TRANSCONDUCTANCE CURVE

$$i_D = 2\ mA\Big|_{v_{GS}=-2\ V}$$

$$v_{GS} = -1.2\ V\Big|_{i_D=4\ mA}$$



LABORATORY